

Transformations of $f(x) = |x|$, Part I

TEACHER KEY | Algebra II | Unit 2 | Day 2 | TEK A2.6.C

Objective: I can describe and graph the effect of a and b on $f(x) = a|bx|$ - vertical/horizontal stretches, compressions, and reflections.

Vocabulary

Transformation — a change to a parent function's graph: **stretch, compression, reflection, or shift**.

Vertical stretch/compression — $af(x)$; $|a| > 1$ **stretches** away from the x-axis, $0 < |a| < 1$ **compresses** toward it.

Horizontal stretch/compression — $f(bx)$; $|b| > 1$ **compresses** toward the y-axis, $0 < |b| < 1$ **stretches** away from it.

Reflection — $a < 0$ flips over the **x-axis**; $b < 0$ flips over the **y-axis**.

Part 1 — The Effect of a in $f(x) = a|x|$

Function	a-value	Effect	Vertex
$f(x) = x $	1	<i>parent function</i>	(0, 0)
$f(x) = 2 x $	2	vertical stretch (steeper)	(0, 0)
$f(x) = (1/2) x $	1/2	vertical compression (flatter)	(0, 0)
$f(x) = - x $	-1	reflection over the x-axis (opens down)	(0, 0)

Notice: no matter what a is, the vertex never moves - a(0) is always 0.

You Try. Describe $f(x) = 4|x|$: **vertical stretch** Describe $f(x) = -(1/3)|x|$: **vertical compression, reflected over the x-axis**

Part 2 — The Effect of b in $f(x) = |bx|$

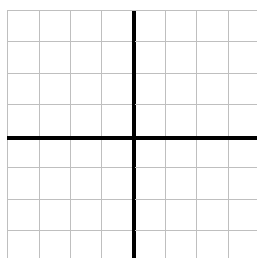
Function	b-value	Effect
$f(x) = x $	1	<i>parent function</i>
$f(x) = 2x $	2	horizontal compression
$f(x) = (1/2)x $	1/2	horizontal stretch
$f(x) = -x $	-1	reflect over the y-axis - but looks IDENTICAL to the parent, since x is already symmetric there

You Try. Describe $f(x) = |3x|$: **horizontal compression** Describe $f(x) = |(1/5)x|$: **horizontal stretch**

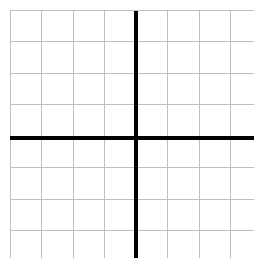
Part 3 — Graph and Compare

Graph both on their own grid. Each square is 1 unit; both axes run -5 to 5. Which one looks steeper?

A. $f(x) = 2|x|$



B. $f(x) = (1/2)|x|$



Steeper graph: **A, since $|a| = 2 > 1$ stretches, while B compresses**

Part 4 — Rank Them

Order from FLATTEST to STEEPEST: $|x|$, $3|x|$, $(1/4)|x|$, $-2|x|$

$(1/4)|x|$, $|x|$, $-2|x|$, $3|x|$ (compare $|a|$: 1/4, 1, 2, 3)

Practice

- Describe the transformation of $f(x) = 5|x|$ from the parent. **vertical stretch by a factor of 5**
- Describe the transformation of $f(x) = |(1/4)x|$ from the parent. **horizontal stretch**
- Describe the transformation of $f(x) = -6|x|$ from the parent. **vertical stretch by 6, reflected over the x-axis**
- Which of these two graphs is steeper: $f(x) = 3|x|$ or $f(x) = (1/5)|x|$? **$f(x) = 3|x|$, since $|a| = 3 > 1$ stretches**
- $f(x) = -4|2x|$ combines an a-effect and a b-effect. Describe both. **vertical stretch by 4 with a reflection over the x-axis, AND horizontal compression**
- True or False: $f(x) = |-5x|$ looks different from $f(x) = 5|x|$. Explain. **False - $|-5x| = |-5||x| = 5|x|$, the same function**
- Describe the transformation of $f(x) = (1/3)|-x|$ from the parent. **vertical compression by 1/3 - the b = -1 reflection over the y-axis is invisible**
- Match each function to its effect: $f(x) = 7|x|$, $f(x) = |7x|$, $f(x) = -|x|$, $f(x) = (1/7)|x|$.
A. vertical compression B. reflection only C. vertical stretch by 7 D. horizontal compression
 $f(x) = 7|x|$: **C** $f(x) = |7x|$: **D** $f(x) = -|x|$: **B** $f(x) = (1/7)|x|$: **A**
- Two functions, $f(x) = a|x|$ and $g(x) = |bx|$, look IDENTICAL for every x you try. What must be true of a and b? **$a = |b|$**
- Describe the transformation of $f(x) = -(2/3)|x|$, using both the stretch/compression amount and the reflection. **vertical compression by 2/3, reflected over the x-axis**

Exit Ticket

Describe every transformation in $f(x) = -3|x|$ from the parent function (stretch/compression amount, and any reflection).

vertical stretch by a factor of 3, reflected over the x-axis (opens downward)